

OpenStack dashboard by navigating to `http://<your-server-ip>/dashboard` in your web browser. This will direct you to the OpenStack login page, as shown in Figure 3.

Creating a virtual machine on the deployed OpenStack setup

Create router with the required parameters: After logging into the OpenStack dashboard, click on `network> routers>create a router`. As shown in Figure 4, provide the router name (OSFY Testing Gateway) and select the external network.

Configure the network: In this step, configure the network, subnet, and IP address for the deployed VM instance. In the subnet tab, provide a subnet name, network address, and gateway IP.

Launch instance: Finally, from the `compute->instance` option, launch the instance. After some time, the deployed instance will start running, as shown in Figure 5.

OpenStack is a popular choice for building private clouds due to its open source nature, flexibility, and modular

architecture. It is widely used by organisations and cloud service providers to build and manage private, public, and hybrid clouds, as it offers a flexible and extensible solution for building scalable and interoperable cloud computing environments. **END** 

References

- <https://www.openstack.org/software/project-navigator/openstack-components#openstack-services>
- <https://cloud.denbi.de/wiki/Concept/openstack/>
- <https://www.openstack.org/marketplace/hosted-private-clouds/>

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desc

The `desc()` function is used to sort a column in the descending order. The bank balances can be displayed from the highest to the lowest values, as shown below:

```
> bank %>% arrange(desc(balance))
  age  job  marital education default balance housing loan
1   60 retired married primary no    71188 no    no
2   42 entrepreneur married tertiary no 42045 no    no
3   43 technician single tertiary no 27733 yes no
4   36 management married tertiary no 27359 yes no
5   57 technician married tertiary no 27069 no yes
6   31 housemaid single primary no 26965 no
...

```

nth

A specific row from the data set can be retrieved using the `nth()` function. The 10th entry in the bank data frame is shown below:

```
> bank %>% nth(10)
  age  job  marital education default balance housing loan
1   43 services married primary no    -88 yes yes
cellular 17
  month duration campaign pdays previous poutcome y
1   apr      313      1   147      2 failure no

```

You are encouraged to read the dplyr package manual to learn more functions, arguments and usage. **END** 

References

- [Moro et al., 2014] S. Moro, P. Cortez and P. Rita; A Data-Driven Approach to Predict the Success of Bank Telemarketing; Decision Support Systems, Elsevier, 62:22-31; June 2014
- Package dplyr; <https://cran.r-project.org/web/packages/dplyr/index.html>

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